



Curriculum

for the Master's degree programme in

Informatics

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Section 1 General Remarks

- (1) The Master's degree programme in Informatics is equivalent to 120 European Credit Transfer System credits (ECTS credits). This equates to an anticipated study duration of four semesters. The Master's degree programme in Informatics is assigned to the group of engineering science degree programmes pursuant to Section 54 (1) of the Universities Act 2002 (UG).
- (2) The workload for the individual study effort is indicated in ECTS credits; the workload for one year should amount to 1,500 full hours, for which 60 ECTS credits are awarded (Section 54 (2) UG). The workload comprises independent study as well as the semester hours/contact hours, including participation in assessment procedures.
- (3) The Master's degree programme in Informatics is taught in English.

Section 2 Qualification Profile and Competences

The competency profile describes the academic and professional qualifications that students gain by completing the degree programme.

- (1) **Initial situation:** The main focus of the Master's degree programme in Informatics at the University of Klagenfurt is on Applied Informatics. The programme aims to qualify graduates for the top-quality development of computer-assisted solutions to problems, for the efficient and effective management of the problem-solving process, and for the design of the beneficial utilization of solutions, based on methods and techniques from informatics. Above all, this requires sound knowledge and expertise in the field of informatics, and a comprehensive understanding of economic interrelations, but it also demands extensive knowledge about possible fields of application. In informatics, software systems are usually developed in organisations and embedded into social systems. Consequently, additional methodological, social and personal skills are necessary for the successful realisation, introduction, and beneficial utilisation of informatics systems.

Furthermore, the field of informatics is characterised by highly dynamic development. The continuous improvement of methods and tools as well as the rapid succession of innovations in the areas of application represent significant challenges in terms of the qualification of graduates. It must be ensured that graduates can actively contribute to this process of improvement and innovation while also being able to update their knowledge in an autonomous and timely manner. This requires not only detailed knowledge about the fundamental concepts, interrelations, and insights gained in the field, upon which the methods and tools are based, but also the reinforcement of individual competencies such as self-organisation and creative drive. A combination of research, teaching and practical application provides ideal conditions for developing these competencies.

- (2) **Skills to be gained:** Building on the initial situation, there are four main fields of competence in which students completing the Master's degree programme in Informatics gain specialised skills and expertise: 1) Specialised knowledge in informatics and in areas of application, 2) Individual and leadership skills, 3) Practical and research experience and problem-solving skills, and 4) Working scientifically. The

skills and expertise to be gained are described below in the form of learning outcomes for each field of competence:

- *Specialised skills in informatics and in areas of application:* Specialist subjects in practical and theoretical informatics serve to expand skills and expertise in the core area of informatics. By choosing a specialist subject - characterised by an emphasis on research-oriented teaching - the students are introduced to the state of the art in terms of research and technology. The specialist subjects enable students to create and implement their own solutions and work scientifically in their chosen area as part of the Master's degree programme. If necessary, students can complete courses in supplementary subjects to gain specific relevant knowledge not previously acquired.
 - *Individual and leadership skills:* Skills consolidation courses in particular prepare students not only for working scientifically, but also for predicting the consequences of technology. The skills and expertise gained allow students to research information autonomously, summarise that information in academic papers and present it. With these skills, students can critically question scientific findings, technical solutions and their impact on society. Gender knowledge and gender competencies can be acquired by attending courses in the elective subject of Feminist Science/Gender Studies (Section 5 (3)).
 - *Practical or research experience and problem-solving skills:* Students have the opportunity to work on a project to gain in-depth project experience. Thus, they can gain intensive practical experience or impressions and experiences in the industrial or research projects in which they are involved. In this way, graduates boost their skills for solving a wide range of problems in business or administration, or when addressing research questions in the academic sphere.
 - *Working scientifically:* Students deliver their first scientific achievements in the seminars for the specialist subject and the Master's thesis, and are thus introduced to working scientifically. The skills and expertise they gain enable students to research a subject autonomously, develop and evaluate solutions for research and practical questions, document the findings in an academic paper and present and discuss them.
- (3) **Professional fields and fields of activity:** The graduates of the Master's degree programme in Informatics are experts in their specialist field with a comprehensive background in informatics. They are able to lead complex software development projects in small, medium-sized and large institutions, define novel problem solutions themselves and further develop the concepts, methods and tools of informatics or one of the areas of application. Following three years of relevant professional activity and the successful completion of the Civil Engineer examination, graduates of the Master's degree programme are entitled to work as chartered civil engineers in the field of informatics. The Master's degree programme also prepares graduates for a doctoral programme in informatics or computer science.

Section 3 Admission Requirements

- (1) Admission to a Master's degree programme is contingent upon completion of a relevant Bachelor's degree programme, another relevant degree programme of at least the

same higher education level completed at a recognised domestic or foreign post-secondary educational institution, or a degree programme as defined in the curriculum of the Master's degree programme (Section 64 (3) UG).

- (2) The Bachelor's degree programme in Applied Informatics at the University of Klagenfurt, the Bachelor's degree programme in Informatics at the Universities of Vienna, Innsbruck, Linz, Salzburg and Graz University of Technology and the Bachelor's degree programme in Software & Information Engineering at Vienna University of Technology are classed as relevant in any case.
- (3) Students who have completed a Bachelor's degree programme in engineering science or an equivalent degree programme at another Austrian or foreign university, university of applied sciences or other recognised post-secondary educational institution not covered by para. 1 or para. 2, are admitted to the Master's degree programme if they provide evidence of at least 75 ECTS credits from the core subjects for informatics. The following are examples of core subjects for informatics:
 - Principles of applied informatics and software development
 - Technical principles and system software
 - Theoretical informatics
 - Software engineering
 - Databases and information systems
 - Artificial intelligence, machine vision, language processing, pattern recognition
 - Multimedia systems
 - Computer architectures, computer networks, embedded systems
 - Information and systems security and human-machine interaction
- (4) In order to compensate for substantial differences in terms of the content of other relevant programmes of at least the same level of higher education completed at a recognised domestic or foreign post-secondary educational institution, supplementary examinations may be prescribed, which must be completed by the end of the second semester of the Master's programme. The Rectorate may determine which of these supplementary examinations are prerequisites for sitting the examinations stipulated in the curriculum of the Master's degree programme (Section 64 (3) UG).
- (5) The provisions governing the two study variants I or II in accordance with para. 5 are defined as follows:
 - (a) Graduates of programmes/degree programmes as defined in para. 4 with prescribed examinations in the supplementary subject must complete study variant II.
 - (b) Graduates of programmes/degree programmes as defined in paras. 1, 2, 3 or 4 who have completed a relevant practical assignment as part of the programme must complete study variant II.
 - (c) All other admitted students can choose between study variant I and II.
- (6) Classification according to para. 5 and the supplementary subject which may need to be taken (Section 10) will be determined by the Programme Director following admission.

Section 4 Academic Degree

Graduates of this Master's degree programme will be awarded the academic title "Diplom-Ingenieurin/Diplom-Ingenieur" (graduate of civil engineering, informatics; shortened to "Dipl.-Ing." or "DI"). If this title has been awarded, it must be placed before the graduate's name.

Section 5 Structure & Organisation of the Degree Programme/Intended Learning Outcomes

- (1) The required subjects, the specialist subject (elective) and the options must be completed as part of the Master's degree programme in Informatics. Two study variants are defined in relation to the completion of a project and/or a supplementary subject. Depending on the classification under Section 3 (5), these can be selected by the student or are specified:
- Study variant I: Project (incl. review)
 - Study variant II: Supplementary subject.

In addition, a Master's thesis must be written and the associated research seminar must be completed. Table 1 shows the structure of the Master's degree programme in Informatics.

- (2) The intended learning outcomes listed in the table below refer to methods that correspond to the current state of the art.

Table 1: Structure of the Informatics Master's degree programme

Subject	Designation	Intended learning outcomes	ECTS credits
Required subjects	1 Specialization in informatics	Following successful completion of this subject, students will be able to: - Explain fundamental and advanced concepts and technology in the fields of data engineering, distributed computing infrastructures, software engineering, artificial intelligence and machine learning and compiler construction. - Devise, implement, test and measure complex programmes and distributed systems according to a prescribed or self-defined process using modern tools and frameworks. - Make profitable use of methods and technology from data engineering, artificial intelligence and machine learning to develop solutions and to make informed decisions in relation to controlling decision-making processes.	22

	2 Advanced Social and Academic Skills	Following successful completion of this subject, students will be able to: - Explain the structure of an academic paper and analyse and discuss it using example articles. - Explain methods for predicting the consequences of technical solutions and apply them to practical examples. - Explain the interrelationships in the fields of gender, technology and science in informatics and critically question them.	12
Guided elective - choice of one area of specialisation	3 Artificial Intelligence	Students know the essential concepts and methods of artificial intelligence, particularly in the fields of knowledge representation and processing, and machine learning and can put these into practice in appropriate areas of application, such as - Automatic planning - Recommendation systems - Semantic web - Diagnostics - Configuration - Verification - Decision support systems - Natural language processing - Cognitive robotics	32
	4 Business Information Systems	Following successful completion of this subject, students will be able to: - Explain fundamental concepts and principles of information systems for recommendation and decision-making processes and apply the associated methods, technologies and systems. - Manage small-scale IT projects and introduce IT management methods in the business world. - Determine, devise, document and implement business processes. - Devise, select and implement e-commerce systems.	
	5 Data Science and Engineering	Following successful completion of this subject, students will be able to: - Devise, implement and optimise large-scale data pools (databases, data warehouses, data lakes). - Integrate heterogeneous data pools. - Select methods for evaluating large quantities of data (in particular data mining processes) for areas of interest; implement and apply them to modern architectures. - Analyse and guarantee data quality.	

	6 Distributed Systems	Following successful completion of this subject, students will be able to: - Explain and apply fundamental concepts, architecture principles, and types of organisation and communication for modern distributed systems. - Localise and analyse a range of problems of distributed systems and devise creative solutions for these problems.
	7 Human-Computer Interaction	Following successful completion of this subject, students will be able to: - Describe and apply fundamental methods and tools for analysing usability/user experiences (including empirical approaches, in particular). - Design and develop classic graphical user interfaces (desktop-based or mobile). - Design and develop non-classical user interfaces (gestures, haptics, voice recognition and synthesis, etc.), determine and understand usage scenarios and quality criteria, and devise and develop prototypical solutions from at least one of these categories.
	8 Information and System Security	Following successful completion of this subject, students will be able to: - Explain the role of IT security as well as IT security combined with risk management along with various security infrastructures (types, structure, purpose, etc.). - Explain basic fundamental mechanisms of cryptology, along with their effects and interaction. - Evaluate threats and associated risks from different perspectives (software, hardware, network, algorithmic, etc.) and identify and implement appropriate countermeasures.

	9 Multimedia Systems	Following successful completion of this subject, students will be able to: - Explain and apply the essential concepts, mechanisms, methods, protocols and components of (distributed) multimedia systems and services. - Apply the fundamental methods and tools for the development of (distributed) multimedia systems and services and use them for the development of simple to moderately complex systems. - Comprehend and analyse simple to moderately complex problems, the recording, processing (incl. compression), communication, storage, analysis, search and retrieval, and high-quality representation of multimedia data and develop solutions for this purpose.	
	10 Software Engineering	Following successful completion of this subject, students will be able to: - Explain reengineering concepts, methods and techniques for analysing and continuously improving the internal quality of complex software systems and apply them to small and medium-size software systems. - Explain the methods and techniques of formal methods and software testing for improving the external quality of complex software systems and apply them to small and medium-size software systems. - Explain the concepts, methods and techniques of continuous integration/continuous delivery for developing complex software systems and apply them to small and medium-size projects.	
Open electives	11 Optional Subjects	Students acquire further individually selected skills.	6
Project	12 Study Variant I: Project (incl. Cross-Project Review)	Following successful completion of this subject, students will be able to: - Define, plan, carry out, document and present a specified area of interest from the field of industry or research according to the rules of well-founded engineering work. - Reflect upon the solution process and the solution to make potential improvements in future.	18

or Supplementary subject	13 Study Variant II: Supplementary Subject	Following successful completion of this subject, students will be able to: After successfully completing the subject, students will be able to explain and apply in-depth knowledge and methods from the selected STEM fields (mathematics, computer science, natural sciences, technology) in accordance with the selected courses.	
Master's thesis	14 Master's Thesis	Following successful completion of this subject, students will be able to: - Research and summarise the latest scientific and technological knowledge in a sub-area of informatics and its applications. - Apply the knowledge gained to design and implement a solution for a problem. - Validate the solution and compare it with other solutions. - Describe, present and discuss a problem, its solution and the results of the validation in the Master's thesis.	25
Research seminar	15 Research seminar		2
Final overall examination	16 Final overall examination		3
Total:			120

- (3) Observe Section 14 for Study Variant I, Project incl. Review, and Section 10 for Study Variant II and the form of the supplementary subject.

Section 6 Study-Related Period Abroad/Mobility

- (1) It is highly recommended for all students of the Master's degree programme to complete a study-related period abroad within the framework of their degree programme. Transnational EU, state or university mobility programmes can be used for this purpose. Preferably, existing offers such as established Erasmus+ or joint study programmes in the field of Informatics should be considered. The 2nd or 3rd semester is the recommended mobility window.
- (2) At the request of regular students who wish to carry out parts of their studies abroad, a preliminary decision shall be issued in advance to determine which of the proposed

examinations and other academic achievements can be recognised (Section 78 (5) UG). In any case, interested students are advised to contact the respective competent Programme Director in advance to discuss possible and intended recognition.

Section 7 Types of Courses

- (1) Lectures (Vorlesungen/VO) are courses by which knowledge is transferred by means of talks given by lecturers. The examination takes place as a one-off (written and/or oral) examination.
- (2) Courses with ongoing assessment are courses in which the assessment does not take place in a one-off examination, but on the basis of written and/or oral contributions by the participants. If, in the framework of a course with ongoing assessment, a seminar paper or a paper requiring a comparable degree of effort is to be written, papers for courses taking place in the winter semester can be handed in up until the following 30 June; papers for courses taking place in the summer semester can be handed in up until 31 January of the following year.

Courses with ongoing assessment comprise:

- (a) Exercise class (Übung/UE): Exercise classes involve carrying out specific tasks in order to consolidate what has been learnt in a lecture.
 - (b) Lecture with workshop (Vorlesung mit Kurs/VC): This course consists of a lecture component and a workshop component. These have common taught content and are assessed together.
 - (c) Practical placement (Praktikum/PR): In addition to scientific education and professional education and training, practical work contributes to the practical and professional objectives of the degree programme. Particular emphasis is placed on working on real tasks and projects. A project study is practical work in which small, applied research or development work is carried out in consideration of all necessary work steps, ideally as a team. A written assignment documenting the course of the project and its results is an inherent part of a project study.
 - (d) Seminar (Seminar/SE): Seminars are used to reflect upon and discuss special scientific problems and/or work. Students make their own oral and written contributions, whereby written work must exhibit an independent academic character both in terms of form and content.
 - (e) Research seminar (Privatissimum/PV): The purpose of the research seminar is to provide ongoing supervision and quality assurance to students as they write their Master's thesis.
- (3) In terms of courses from other curricula, the definitions found in the respective other curricula apply.

Section 8 Required Subjects

- (1) Required subjects are subjects significant to the degree programme and for which examinations must be taken. Students must complete a total of 34 ECTS credits in the required subjects. The semester hours (S.h.) listed are recommendations for the purpose of planning and organising courses.
- (2) The courses for the required subjects can be found in **Table 2** below.

Table 2: Required subjects and assigned courses

	Course code	Course type	S.h.	ECTS credits
Informatics specialist course	1.1 Data Engineering	VC	2	4
	1.2 Distributed Computing Infrastructures	VC	2	4
	1.3 Advanced Software Engineering	VC	2	4
	1.4 Artificial Intelligence & Machine Learning	VC	2	4
	1.5 Compiler Construction	VO + UE	2 + 2	2 + 4
Total:				22
Skills consolidation	2.1 Scientific Writing	SE	2	4
	2.2 Technology Assessment	VC	2	4
	2.3 Selection of one of the following courses - 2.3.1 Reflecting on the Limits of Formal Sciences - 2.3.2 Course from the elective programme in Feminist Science/Gender Studies			4
Total:				12

Section 9 Guided Electives

- (1) Guided electives are those subjects that students choose according to the provisions of the curriculum. A total of 32 ECTS credits must be completed in guided electives. The semester hours (S.h.) listed are recommendations for the purpose of planning and organising courses.
- (2) Electives serve to extend knowledge, methods and skills in the selected area and are therefore regarded as and called specialist subjects. Students must select one of the following subjects as a specialist subject:
 - Artificial Intelligence
 - Business Information Systems
 - Data Science and Engineering
 - Distributed Systems
 - Human-Computer Interaction
 - Information and System Security
 - Multimedia Systems
 - Software Engineering

Table 3 shows the structure of each specialist subject. In the selected specialist subject, courses totalling at least 16 ECTS credits are to be completed, including exactly 1 seminar in the specialisation totalling 4 ECTS credits.

- (3) Students can complete further courses totalling 16 ECTS credits from the selected or other specialisations. Note that only one seminar (which is to be completed according to para. 2 in the selected specialisation) is taken into account for the electives in the Master's degree programme.

Table 3: Structure of a specialist subject (X denotes the selected specialist subject according to Para. 2)

	Course code	Course type	S.h.	ECTS credits
Specialist subject	X.1 seminar	SE	2	4
	X.2 Further courses from the selected specialist subject			12
	X.3 Further courses from the selected specialist subject or other specialist subjects	VC	2	16
			Total:	32

Section 10 Supplementary Subject

Depending on a student's classification according to Section 3 (5), a supplementary subject can or must be completed (Study Variant II according to Section 5). Students can choose any Bachelor's and/or Master's courses from the STEM fields (mathematics, computer science, natural sciences, technology) totalling 18 ECTS credits. Note that the supplementary subject courses must always be determined in advance with the Programme Director.

Section 11 Open Electives

- (1) Open electives are those subjects that students can freely choose from the range of courses offered by recognised domestic and foreign universities. Courses completed as a prerequisite to study or to gain general or special eligibility for university admission cannot be used for the open electives.
- (2) Students are required to complete 6 ECTS credits in open electives.
- (3) In the case of courses that have been completed at other recognised Austrian or international post-secondary educational institutions, the responsible university body will decide whether recognition as an open elective for the chosen programme of study makes sense academically or with regard to professional activities.

Section 12 Courses with a Limited Number of Participants

- (1) The maximum number of participants permitted on each of the following courses is as follows:
 - Exercise class (UE): 30
 - Lecture with workshop (VC): 30

- Practical placement (PR): 15
- Seminar (SE): 15
- Research seminar (PV): 15

For courses from other curricula, the maximum numbers found in the respective curricula apply.

- (2) If the number of applications for these courses exceeds the number of places available, students will be accepted in accordance with the following procedure:
 - (a) Students for whom the course is a required subject or elective in their curriculum are given priority.
 - (b) If the number of applications still exceeds the number of available places, the students are ordered based on the already acquired ECTS credits for the curriculum that stipulates the course in question as a required subject/ elective. A higher total is ranked higher. If the number of ECTS credits is the same, students will be drawn at random.
 - (c) Up to 10% of the places available will be allocated on a priority basis to students completing a portion of their programme of study at University of Klagenfurt as part of a mobility programme.
- (3) Depending on the financial resources available, parallel courses may be offered for the course in question.

Section 13 Master's Thesis

- (1) The Master's thesis is the academic paper that demonstrates the student's ability to achieve adequate standards of content and methodology when independently addressing scholarly topics. The assignment for the Master's thesis will be so chosen that it is reasonable to expect a student to complete it within six months. A number of students may jointly address a topic, provided that the performance of individual students can be assessed.
- (2) The topic to be covered in the Master's thesis must be chosen from one of required subjects or electives.
- (3) The Master's thesis comprises 25 ECTS credits. In relation to the Master's thesis, the research seminar (2 ECTS credits) must be completed with the Master's thesis supervisor.
- (4) Pursuant to Part B, Section 18 (4) and (2a) of the Statute, the topic and the Master's thesis supervisor must be approved by the Rector of Studies. The application must be made prior to starting work. The student is permitted to change supervisor until the time that the Master's thesis is submitted. Supervision by two persons who are authorised to supervise shall be permissible on a case-by-case basis, where there is good reason to do so (interdisciplinary focus of the topic).
- (5) The completed Master's thesis must be submitted to the Rector of Studies in electronic format. On the request of the supervisor, the author must provide them with a bound copy of the thesis. The supervisor will have two months from the date of submission to assess the Master's thesis.

Section 14 Provisions Regarding the Completion of a Relevant Internship

- (1) During the course of the Master's degree programme - under the conditions of Section 3 (4) and Sections 5, 9 and 10 - relevant work experience, subsequently referred to as a project, can be completed to test and apply in practice and extend the knowledge and skills students have gained. The internship must last at least 350 hours. Students are also required to prepare an internship report. 17 ECTS credits are allocated to the internship including the internship report.
- (2) The project is a guided project study, supervised by a university teacher, in which work is carried out on a self-contained, pre-defined project. The latter applies in particular to working students. Students can suggest the topic/assignment of the project from the specialist subjects (Section 9) and the institution for the project or select them from a list of suggestions. Before beginning the project, approval must be obtained from the supervisor. If necessary due to geographical distance, students are supervised by means of technical tools for communication and collaboration, so that the teaching objectives of the project can be achieved.
- (3) A written report to document the content, results and experiences must be drawn up as part of the project. The supervising university teacher evaluates the project on the basis of this report and a discussion.
- (4) Following the project, normally in the same or following semester, students must present the content of the written report in a presentation as part of a joint course "Cross Project Review" (1 ECTS credit). The "Cross-Project Review" course is evaluated separately from the project.

Section 15 Use of Languages Other than English

As a rule, the courses and oral and written examinations for the Master's degree programme in Informatics will be held and taken in English; the Master's thesis must be written in English.

Section 16 Examination Regulations

- (1) Course examinations for lectures (VO) - preferably in written form - are taken at the end of or after the course as a single examination and cover the course material. The purpose is to assess how successfully students have participated in the course and demonstrate the students' mastery of the knowledge, methods and skills gained in the lecture. In particular, the educational objectives defined in the competency profile (Section 2) are to be used as a benchmark.
- (2) All other types of course have ongoing assessment; attendance is compulsory. Exercise classes (UE) and practical work (PR) are assessed through accompanying checks and also through written and oral examinations as well as on the basis of practical activities. Students' written and oral contributions (especially seminar work, seminar talks and participation in discussions) are used as a benchmark for assessment in seminars (SE) and research seminars (PV). In a lecture with workshop (VC), the mode of examination is to be determined based on the character of the course and the educational objectives.

- (3) The course instructors must provide students with information about the course objectives, contents, methods and the criteria of course assessment and examinations prior to the start of each semester.
- (4) Examinations and other academic achievements that have already been counted in order to complete the degree programme serving as prerequisite for admission cannot be re-used in this Master's degree programme for the purpose of obtaining the degree.
- (5) To graduate from the Master's degree programme in Informatics, students are required to successfully complete the following course components:
 - (a) The courses for the required subjects, electives and options according to Sections 8, 9 and 11
 - (b) The work experience or the supplementary subject courses according to Sections 10 and 14
 - (c) The Master's thesis and the associated research seminar according to Section 13
 - (d) The final comprehensive oral examination before a committee according to Para. 6
- (6) The prerequisite for applying for the final overall examination before a committee is the successful completion of the parts listed under para. 5, points a-c. The final overall examination before a committee takes place in front of a three-person examination committee and covers:
 - a) The subject that the topic of the Master's thesis falls under (in the form of a presentation and defence of the Master's thesis); 1.5 ECTS credits
 - b) Another subject from the Master's degree programme which has to be selected from the informatics specialist course (Section 8), the electives (Section 9) or, where applicable, the supplementary subject (Section 10), and which is different to point a) in terms of content; 1.5 ECTS credits
- (7) The person supervising the Master's thesis is not permitted to be the chairperson of the examination committee.
- (8) Assessment by means of a grading scheme is stipulated for the relevant internship and the course "Cross Project Review" in accordance with Section 14.
- (9) The regulations of the Statute of the University of Klagenfurt, Part B and the Universities Act as amended from time to time apply to the implementation and repetition of examinations.

Section 17 Effective Validity

- (1) This curriculum will enter into force after announcement in the University of Klagenfurt university bulletin as of 1 October 2019 and will apply to all students who commence their Master's degree programme from the 2019/20 winter semester onwards.
- (2) The non-structural changes to the curriculum, published in bulletin Issue 17, No. 108.3 dated 7 June 2023 of the University of Klagenfurt, will enter into force on 1 October 2023. All students enrolled in the Master's degree programme are subject to the amended curriculum from the time of effective validity.

APPENDIX Non-binding Recommended Course of Study

The following tables suggest how the ECTS credits could be distributed across the semesters.

Non-binding recommended route for Study Variant I according to Section 5 (1):

	1st semester	2nd semester	3rd semester	4th semester
1.1 Data Engineering	4 ECTS credits			
1.2 Distributed Computing Infrastructures	4 ECTS credits			
1.3 Advanced Software Engineering	4 ECTS credits			
1.4 Artificial Intelligence & Machine Learning		4 ECTS credits		
1.5 Compiler Construction		6 ECTS credits		
2.1 Scientific Writing		4 ECTS credits		
2.2 Technology Assessment			4 ECTS credits	
2.3 Selection of one of the following courses - 2.3.1 Reflecting on the Limits of Formal Sciences - 2.3.2 Course from the elective programme in Feminist Science/Gender Studies			4 ECTS credits	
3-10 Elective Subject	12 ECTS credits	16 ECTS credits	4 ECTS credits	
11 Optional Subjects	6 ECTS credits			
12 Project (incl. Cross-Project Review)			18 ECTS credits	
14 Master's thesis (incl. research seminar)				27 ECTS credits
15 Final Overall Examination				3 ECTS credits
Total:	30 ECTS credits	30 ECTS credits	30 ECTS credits	30 ECTS credits

Non-binding recommended route for Study Variant II according to Section 5 (1):

	1st semester	2nd semester	3rd semester	4th semester
1.1 Data Engineering	4 ECTS credits			
1.2 Distributed Computing Infrastructures	4 ECTS credits			
1.3 Advanced Software Engineering	4 ECTS credits			
1.4 Artificial Intelligence & Machine Learning		4 ECTS credits		
1.5 Compiler Construction		6 ECTS credits		
2.1 Scientific Writing		4 ECTS credits		
2.2 Technology Assessment			4 ECTS credits	
2.3 Selection of one of the following courses - 2.3.1 Reflecting on the Limits of Formal Sciences - 2.3.2 Course from the elective programme in Feminist Science/ Gender Studies			4 ECTS credits	
3-10 Elective Subject	8 ECTS credits	8 ECTS credits	16 ECTS credits	
11 Optional Subjects			6 ECTS credits	
13 Supplementary subject	12 ECTS credits	6 ECTS credits		
14 Master's thesis (incl. research seminar)				30 ECTS credits
Total:	32 ECTS credits	28 ECTS credits	30 ECTS credits	30 ECTS credits